

LAB EXPERIMENT RECORD

Department	AIDS
Course Code / Course Name	DSA0601 / Data Handling and Visualization
Experiment	Data Visualization using R
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Day	Tuesday

SET 5: WEBSITE ANALYTICS

Dataset: Website Traffic

Date	Page Views	Click-through Rate
2023-01-01	1500	2.3%
2023-01-02	1600	2.7%
2023-01-03	1400	2.0%
2023-01-04	1650	2.4%
2023-01-05	1800	2.6%

Aim

To analyze website traffic using line, bar, and stacked area charts, and to develop an interactive Tableau dashboard for exploring website traffic data.

PROGRAM 1: Line Chart of Daily Page Views

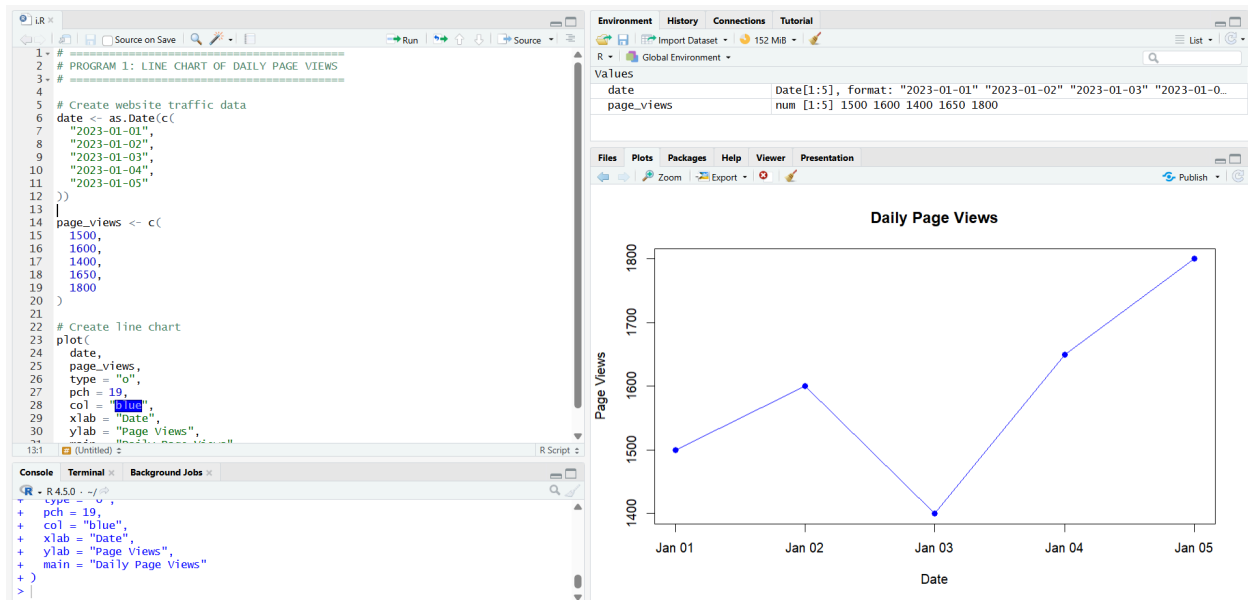
Question

Create a line chart to visualize the trend in daily page views over time. Label the axes and title the chart.

R Program

```
# =====  
# PROGRAM 1: LINE CHART OF DAILY PAGE VIEWS  
# =====  
  
# Create website traffic data  
date <- as.Date(c(  
  "2023-01-01",  
  "2023-01-02",  
  "2023-01-03",  
  "2023-01-04",  
  "2023-01-05"  
))  
  
page_views <- c(  
  1500,  
  1600,  
  1400,  
  1650,  
  1800  
)  
  
# Create line chart  
plot(  
  date,  
  page_views,  
  type = "o",  
  pch = 19,  
  col = "blue",  
  xlab = "Date",  
  ylab = "Page Views",  
  main = "Daily Page Views"  
)
```

Output:



Observation

Page views decreased from 1600 on January 2 to 1400 on January 3. After that, page views increased and reached the highest value of 1800 on January 5.

PROGRAM 2: Bar Chart of Top N Days with Highest Click-through Rates

Question

Generate a bar chart showing the top N days with the highest click-through rates. Label the chart elements.

For this practical, we'll take:

$$N = 3$$

So we will display the **top 3 days**.

Data

Date	Click-through Rate
2023-01-01	2.3%
2023-01-02	2.7%
2023-01-03	2.0%
2023-01-04	2.4%
2023-01-05	2.6%

The top 3 are:

1. January 2 → 2.7%
2. January 5 → 2.6%
3. January 4 → 2.4%

R Program

```
# =====  
# PROGRAM 2: TOP 3 DAYS BY CLICK-THROUGH RATE  
# =====  
  
# Create data  
date <- c(  
  "2023-01-01",  
  "2023-01-02",
```

```
"2023-01-03",  
"2023-01-04",  
"2023-01-05"  
)
```

```
click_rate <- c(  
  2.3,  
  2.7,  
  2.0,  
  2.4,  
  2.6  
)
```

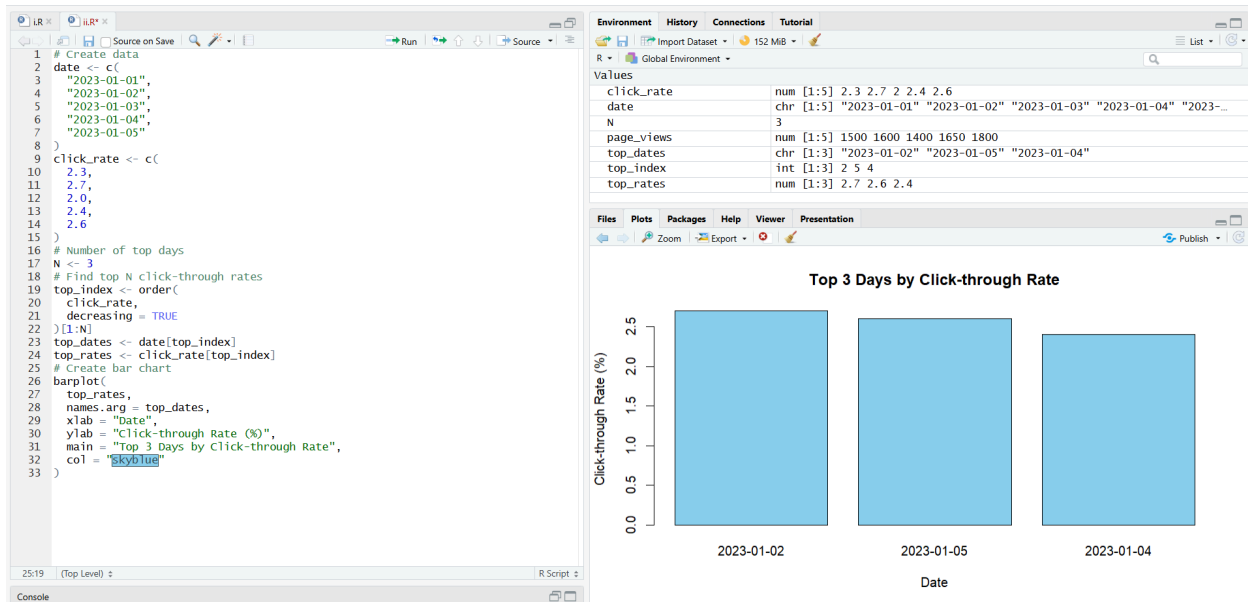
```
# Number of top days  
N <- 3
```

```
# Find top N click-through rates  
top_index <- order(  
  click_rate,  
  decreasing = TRUE  
)[1:N]
```

```
top_dates <- date[top_index]  
top_rates <- click_rate[top_index]
```

```
# Create bar chart  
barplot(  
  top_rates,  
  names.arg = top_dates,  
  xlab = "Date",  
  ylab = "Click-through Rate (%)",  
  main = "Top 3 Days by Click-through Rate",  
  col = "skyblue"  
)
```

Output:



Observation

January 2 has the highest click-through rate of 2.7%, followed by January 5 with 2.6% and January 4 with 2.4%.

PROGRAM 3: Stacked Area Chart of User Interactions

Question

Develop a stacked area chart to display the distribution of user interactions such as likes, shares, and comments on a website.

Sample data:

Date	Likes	Shares	Comments
2023-01-01	120	40	20
2023-01-02	150	50	25
2023-01-03	100	35	18
2023-01-04	160	55	30
2023-01-05	180	65	35

R Program

```
# =====  
# PROGRAM 3: STACKED AREA CHART  
# =====  
  
# Date  
date <- as.Date(c(  
  "2023-01-01",  
  "2023-01-02",  
  "2023-01-03",  
  "2023-01-04",  
  "2023-01-05"  
))  
  
# Sample user interaction data  
likes <- c(  
  120,  
  150,  
  100,  
  160,  
  180
```

)

```
shares <- c(  
  40,  
  50,  
  35,  
  55,  
  65  
)
```

```
comments <- c(  
  20,  
  25,  
  18,  
  30,  
  35  
)
```

Calculate cumulative values

```
total_likes <- likes
```

```
total_shares <- likes + shares
```

```
total_comments <- likes + shares + comments
```

Create empty plot

```
plot(  
  date,  
  total_comments,  
  type = "n",  
  ylim = c(0, max(total_comments)),  
  xlab = "Date",  
  ylab = "Number of Interactions",  
  main = "Website User Interactions"  
)
```

Likes area

```
polygon(  
  c(date, rev(date)),  
  c(  

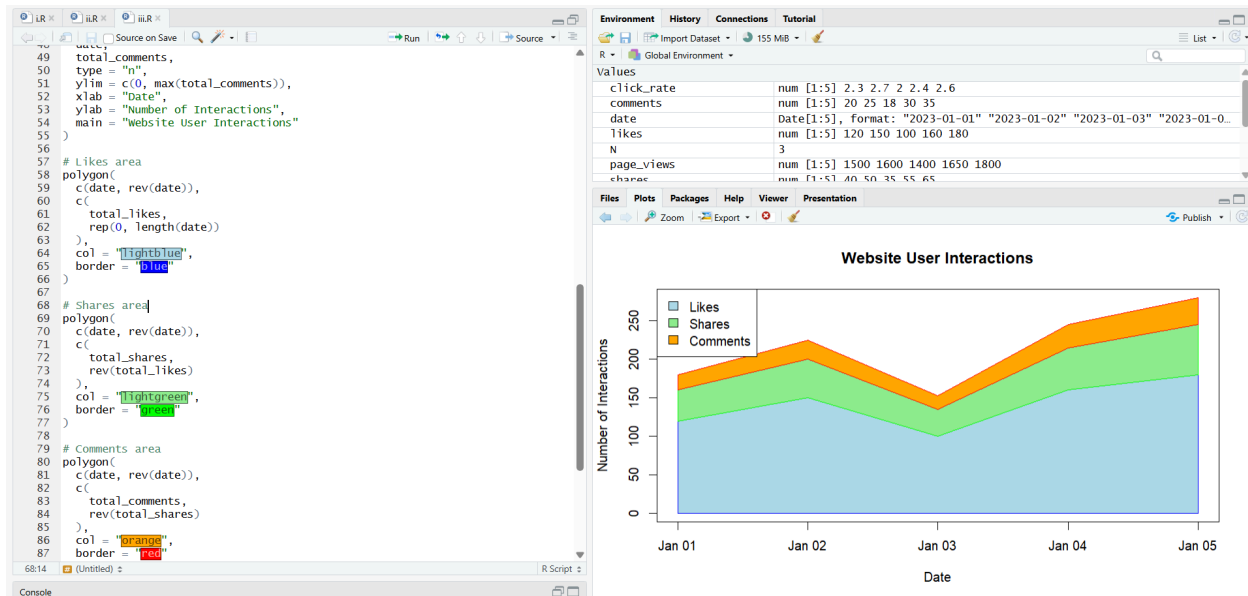
```

```

    total_likes,
    rep(0, length(date))
  ),
  col = "lightblue",
  border = "blue"
)
# Shares area
polygon(
  c(date, rev(date)),
  c(
    total_shares,
    rev(total_likes)
  ),
  col = "lightgreen",
  border = "green"
)
# Comments area
polygon(
  c(date, rev(date)),
  c(
    total_comments,
    rev(total_shares)
  ),
  col = "orange",
  border = "red"
)
# Add legend
legend(
  "topleft",
  legend = c(
    "Likes",
    "Shares",
    "Comments"
  ),
  fill = c(
    "lightblue",
    "lightgreen",
    "orange"
  )
)
)

```

Output:



Observation

The stacked area chart shows how likes, shares, and comments contribute to total website interactions over time. The total interaction level generally increases toward January 5.

PROGRAM 4: Tableau Dashboard

Question

Build a Tableau dashboard with the line chart and bar chart for interactive exploration of website traffic data.

Step 1: Prepare the Excel Dataset

Create an Excel file with:

Date	Page Views	Click-through Rate
2023-01-01	1500	2.3
2023-01-02	1600	2.7
2023-01-03	1400	2.0
2023-01-04	1650	2.4
2023-01-05	1800	2.6

Save it as:

Website_Traffic.xlsx

Tip: In Excel, enter the click-through rate as 2.3, 2.7, etc. and format that column as Percentage only if you want Tableau to display it as a percentage. Otherwise, keeping it as 2.3, 2.7, etc. makes the values represent percentage points.

Step 2: Open Tableau

Open **Tableau Desktop** or **Tableau Public**.

Select:

Connect → Microsoft Excel

Then select:

Website_Traffic.xlsx

Step 3: Create the Line Chart

Create **Sheet 1**.

Drag:

Date



Columns

Drag:

Page Views



Rows

Select:

Marks → Line

Give the worksheet the title:

Daily Page Views

You will get a line chart showing page views from January 1 to January 5.

Step 4: Create the Bar Chart

Create a **new worksheet**.

Drag:

Date



Columns

Drag:

Click-through Rate



Rows

Select:

Marks → Bar

To show only the **Top 3 days**, use a Top N filter.

Go to:

Date → Filter → Top → By field

Set:

Top 3

by

Click-through Rate

Give the worksheet the title:

Top 3 Days by Click-through Rate

Step 5: Create the Dashboard

Click:

New Dashboard

Add both worksheets:

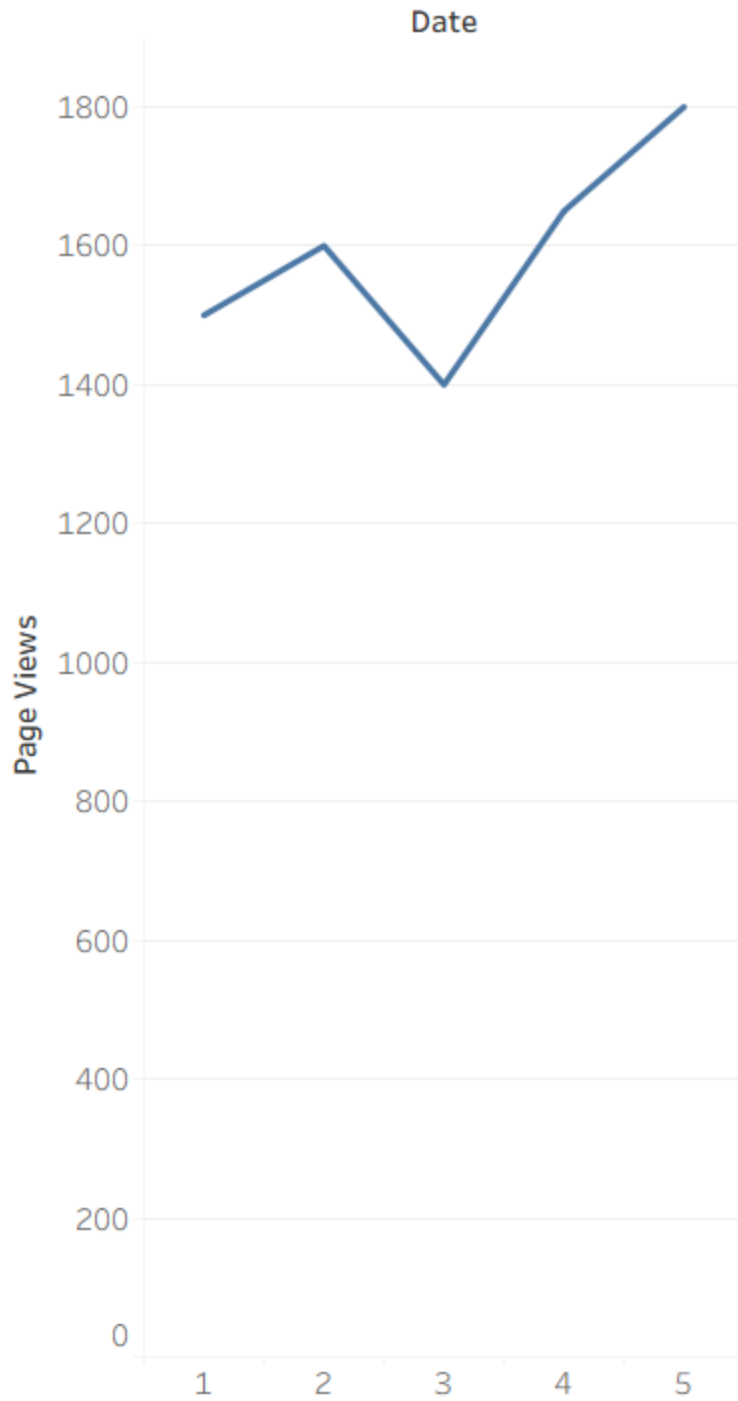
Columns

DAY(Date)

Rows

SUM(Page Views)

Daily Page Views



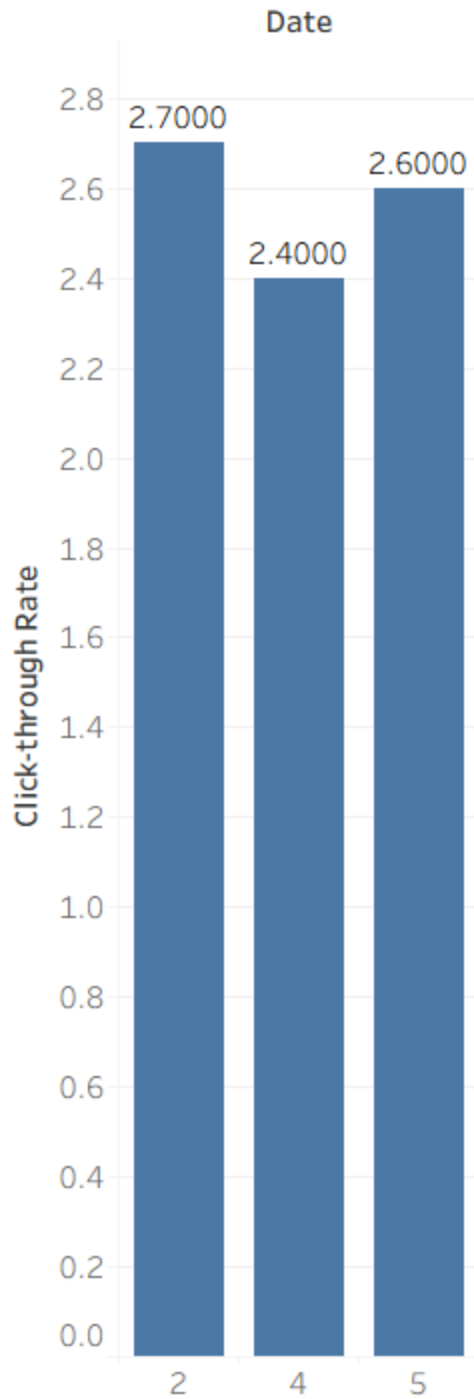
Columns

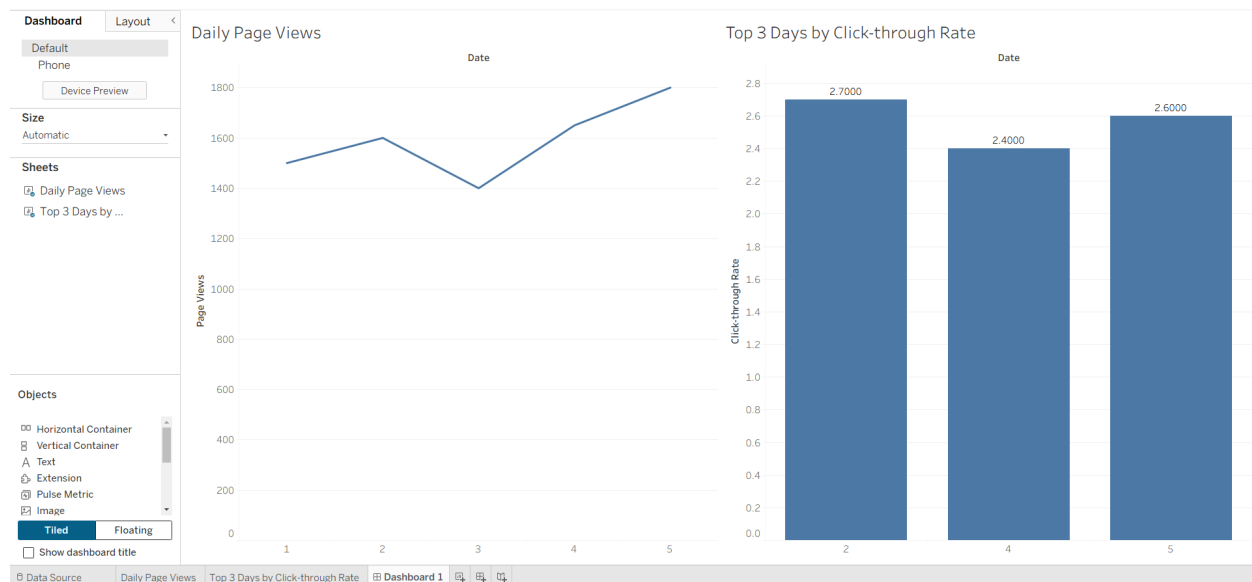
DAY(Date)

Rows

SUM(Click-through ..

Top 3 Days by Click-through Rate





Observation

The Tableau dashboard combines the daily page-view line chart and the top 3 click-through-rate bar chart. Users can interact with the dashboard to explore website traffic trends and identify days with higher click-through rates.

Procedure

1. Create the website traffic dataset in R.
2. Convert the date values into Date format.
3. Use `plot()` to create the daily page-view line chart.
4. Use `order()` to identify the top N click-through-rate days.
5. Use `barplot()` to create the Top 3 click-through-rate bar chart.
6. Prepare sample likes, shares, and comments data.
7. Calculate cumulative interaction values.
8. Use `polygon()` to construct the stacked area chart.
9. Add a legend to identify likes, shares, and comments.
10. Prepare the website traffic data in Excel.
11. Import the Excel file into Tableau.
12. Create the daily page-view line chart.
13. Create the click-through-rate bar chart.
14. Apply a Top 3 filter.
15. Combine both charts into a Tableau dashboard.
16. Add filters/actions for interactive exploration.

Result

The website traffic data was successfully visualized using a line chart for daily page views, a bar chart for the top 3 click-through-rate days, and a stacked area chart for user interactions. An interactive Tableau dashboard was also developed by combining the line chart and bar chart.